Module 1 :- AWS

Client-Server :- In computing, a client can be a web browser or desktop applications that a person interacts with to make req to Computer Servers.

client → Customer → Request ⇒ (The Request can be anything in the computing world.)

Server → Source

→ Basically a Customer makes a request, & with permissions, the server responds to that request.

In AWS, the server (Services) would be called an Amazon Elastic Compute cloud or EC2, an EC2 instance, a virtual server.

Key Concept to AWS :- "You only pay for what you use".

Cloud Computing :- It is the on-demand delivery of IT resources over the Internet with pay as you go pricing.

On demand delivery indicates that AWS has the resources we need, when we need them. We don't need to tell in advance that we are going to need them.

That kind of flexibility is just not possible when we are managing our own data centres.

The idea of IT resources is actually a big part of the AWS philosophy. We often get asked why AWS has so many products & the answer is :-

Because businesses need them.

(At AWS, we call that the undifferentiated heavy lifting of IT. Tasks that are common, often repetitive, & ultimately time consuming. These are the tasks AWS wants to help us with.

Over the internet, seems simple enough, but it implies that you can access those resources using a secure webpage console or programmatically.)

Pay-as-you-go-pricing.

Deployment Models for cloud computing :—

When selecting a cloud strategy, a company must consider factors :—

(a)
- → Cloud Applications.
- → Components.
- → Preferred Resource Management tools.
- → Any legacy IT infrastructure requirements.

The 3 cloud computing deployments models :—

- → cloud based
- → On premises.
- → Hybrid.

Cloud Based :—

* Run all parts of the Application in the cloud.
* Migrate existing applications to the cloud.
* Design & Build new applications in the cloud.

→ We can build those applications on low-level infrastructure that requires our IT staff to manage them. Alternatively, we can build them using higher-level services that reduce the management, architecting & scaling requirements of the core infrastructure.

On premises :— (Private cloud Deployment)

* Deploy resources by using virtualization & resource management tools.
* Increase resource utilization by using application management & virtualization Technologies

Hybrid Deployment :—

* Connect cloud based resources to on premises infrastructure.
* Integrate cloud based with legacy IT applications.

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Benefits of Cloud Computing:—

(upfront expense → Data centers, physical servers)

→ Trade upfront expense for variable expense.
→ Stop spending money to run & maintain data centers.
→ Stop guessing capacity. (No predictions)
→ Benefit from massive economies of scale.
→ Increase speed & Agility.
→ Go global in Mins

Cost effective → Good results with low cost

Flexible → can be able to change at any condition

(Variable expense → you only pay for computing resources you consume instead of investing heavily in data centers & servers).

How does the scale of cloud computing help you to save costs?

→ The aggregated cloud usage from a large no of customers results in lower pay as you go prices.

⇒ → It describes how customers can benefit from massive economies of scale in cloud computing.

Module 2:— Amazon Elastic Compute Cloud (Amazon EC2):—

→ Generally, we need raw compute capacity to host the applications & provide the compute power that according to the business needs.

→ When we are working with AWS, those servers are virtual & the servers we use to gain access to virtual servers is called EC2.

→ Using EC2 for compute is highly flexible, cost effective & quick when we compare it to running our own servers on premises in the data center that we own.

→ The time & money it takes to get up & running with on-premises resources is fairly high.

EC2 is:—
(It is flexible, Reliable, Scalable)

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Amazon EC2 getting started:—

→ AWS Built Datacenters.
→ AWS Secured Datacenters.
→ AWS purchased Servers. (Physical Servers)
→ AWS Installed it.
→ The Servers are online & ready to use.

(All u have to do is request the EC2 instances you want & they will launch & boot up, ready to be used within a few mins).
(Once you are done, you can easily stop or terminate the EC2 instances).
(Your usage of EC2 instances can vary greatly over time & you only pay for what you use.) Because with EC2, you only pay for running instances, not stopped or terminated instances.

→ EC2 runs on top of physical host machines managed by AWS using Virtualization Technology. When u spin up an EC2 instance, u aren't necessarily taking an entire host to urself. Instead, u r sharing the host with multiple other instances. Otherwise known as virtual machines.
→ And a hypervisor running on the host machine is responsible for sharing the underlying physical resources b/w the virtual machines.
→ This idea of sharing underlying hardware is called Multitenancy.
→ The hypervisor is responsible for coordinating this multitenancy & it is managed by AWS.

Amazon EC2 Configurations:—

→ We can choose the OS :— Based on Windows/Linux
→ Beyond the OS, we can also configure what software we want running on the Instance ⇒ Internal Business Apps, Web Apps, Databases, 3rd Party Softwares.

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EC2 Instances are also resizable. You might start with a small instance, realize the application you are running is starting to max out that server, & then you can give that instance more memory & more CPU → Vertical Scaling.

In essence, you can make instances bigger & or smaller whenever you need to.

→ You also control the networking aspect of EC2.
So what type of requests make it to your server & if they are publicly or privately accessible is something you decide.

→ AWS has just made it much, much easier & more cost effective for you to acquire servers through this Compute as a Service (CaaS).

AWS has different types of EC2 Instances:—
That u can spin up & deploy into ur AWS environment.

Each Amazon EC2 instance type is grouped under an instance family & are optimized for certain types of tasks.

Instance types offer varying combinations of CPU, memory, storage & networking capacity & give you the flexibility to choose the appropriate mix of resources for your applications.

Amazon EC2 instance families:—
→ General Purpose
→ Compute optimized
→ Memory optimized
→ Accelerated Computing
→ Storage optimized.

General purpose provides:—
→ Balanced resources.
→ Diverse workloads.
→ Web servers.
→ Code repositories.

Compute optimized Ideal for:—
→ Compute Intensive Tasks
→ Gaming servers.
→ High Performance Computing (HPC).
→ Scientific Modeling.

Memory optimized are good for:—
→ Memory Intensive tasks

Accelerated Computing good for:—
→ Floating point number calculation.
→ Graphics processing.
→ Data Pattern Matching.
→ Utilize Hardware Accelerators.

Storage optimized:—
→ High performance for locally stored data.

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EC2 pricing:—

Amazon EC2 purchase options:—

→ On-Demand.
→ Savings Plan
→ Reserved Instances.
→ Spot Instances.
→ Dedicated Hosts.

On-Demand:— It means that you only pay for the duration that your Instance runs for. This can be per hour or per second, depending on the instance type & the OS you choose to run. Plus, No long-term commitments / upfront payments are needed. This type of pricing is usually for when you get started & want to spin up servers to test out workloads & play around. You don't need any prior contracts / communications with AWS to use on-demand-pricing. You can also use them to get a baseline for your average usage, which leads us to our next pricing option, Savings plan.

Savings Plan:— It offers low prices on EC2 usage in exchange for a commitment to a consistent amount of usage measured in dollars per hour for a 1 or 3 year term. This flexible pricing model can therefore provide savings of up to 72% on your AWS Compute usage. This can lower prices on your EC2 usage, regardless of instance family, size, OS, tenancy / AWS region. This also applies to AWS Fargate & AWS Lambda usage, which are serverless compute ~~options that~~

Reserved Instances:— These are suited for steady-state workloads or ones with predictable usage & offer you up to a 75% discount versus On-demand pricing. You qualify for a discount once you commit to one/3 year term & can pay for them with 3 payment options:—
All upfront, where you pay for them in full when you commit; & no upfront
partial upfront, where " " " a portion " " "
No upfront, where you don't pay anything at the beginning.

Spot Instances :- They allow you to request spare Amazon EC2 computing capacity for up to 90% off of the On Demand price. The catch here is that AWS can reclaim the instance at any time they need it, giving u a 2 min warning to finish up work & save state. You can always resume later if needed.

Dedicated Hosts :- They are physical hosts dedicated for your use for EC2. These are usually for meeting certain compliance requirements & nobody else will share tenancy of that host.

(Major benefit of AWS :- Scalability & Elasticity)

Scaling Amazon EC2 :- (Backup).

→ Scalability :- The ability of an organization (or a system, such as a computer network) to perform well under an increased / expanding workload.

→ Elasticity :- Ability of a deformed material body to return to its original shape & size when the forces causing the deformation are removed.

Scaling up :- It means adding more power to the machines that are running.

→ Amazon EC2 Auto Scaling :- Adds the instances based on demand & then decommissions instances when they are no longer needed.
This means that every min of the day, we always have the correct no of instances.

Directing traffic with Elastic load Balancing :-

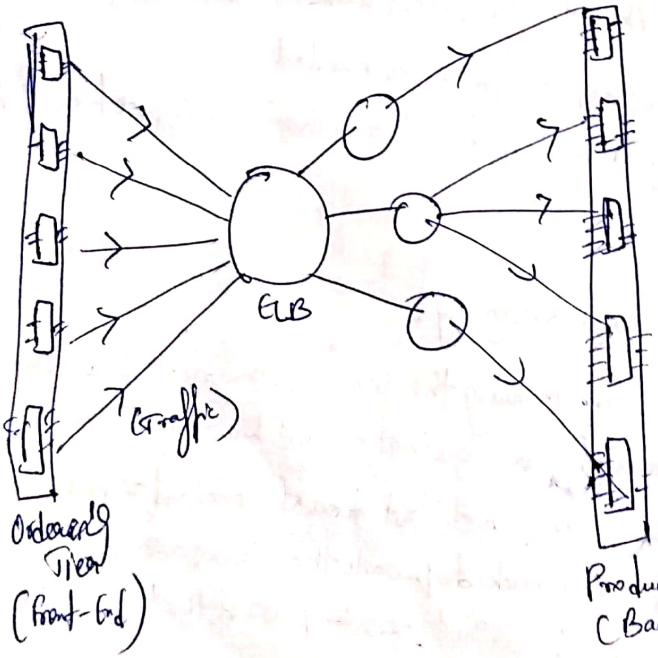
When you have multiple EC2 instances all running the same programs, to serve the same purpose & a request comes in, how does that request know which EC2 instance to go to? How can you ensure there's an even distribution of workload across EC2 instances? So not just one is backed up while the others are idle sitting by. You need a way to route requests to instances to process that request
What u need to solve this is called load Balancing

→ A load Balancer is an application that takes in requests & routes them to the instances to be processed.
Now, there are many off the shelf load balancers that work great on AWS.
In which case it will be up to your operations team to install, manage, update, scale, handle fail over & availability. It's a double.

Properly Distribute Traffic :-

→ High Performance.
→ Cost efficient.
→ Highly Available.
→ Automatically Scalable.

Elastic Load Balancing (ELB) :- It is automatically scalable. As your traffic grows ELB is designed to handle the additional throughput with no change to the Hourly cost. When your EC2 fleet auto scales out, as each instance comes online, the auto scaling service just lets the ELB service know that it's ready to handle the traffic & off it goes. Once the fleet scales in, ELB first stops all new traffic & waits for the existing requests to complete to drain out. Once they do that, then the auto scaling engine can terminate the instances without disruption to existing customers.
ELB is not only used for external traffic.



Messaging & Queuing :-

Applications send msgs to each other to communicate.
If applications communicate directly → then it will be tightly coupled architecture

If a single component fails/changes, it causes issues for the other components or even the whole system

For Ex :- If we have Application A & it is sending msgs directly to Application B, If App B has a failure & can't accept those msgs, App A will begin to see errors as well. This is a tightly coupled architecture.

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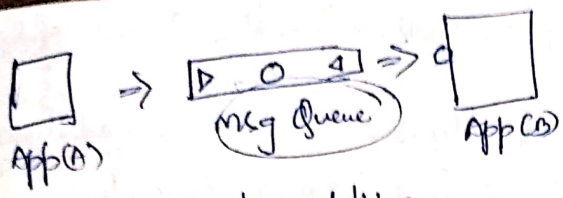
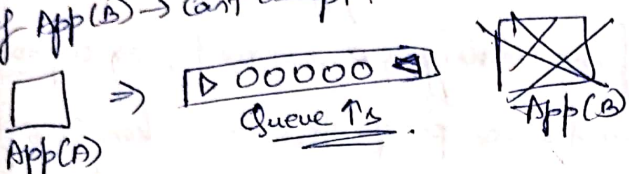
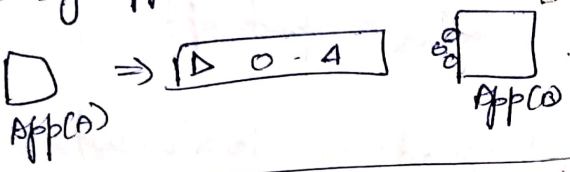


If App(B) → can't accept!—



If App(B) ⇒ Accessed



And this brings 2 AWS Services:-

i) Amazon SQS
(Simple Queue Service)

ii) Amazon SNS
(Simple Notification Service).

Amazon SQS:- It allows us to:-

→ Send Msgs
→ Store Msgs
→ Receive Msgs
} This is without losing msgs or requiring other services to be Available.

Between Software components at any volume.

(Payload:- Data contained with a msg).
↓
(It is protected until delivery).

Amazon SQS queues:- Where messages are placed until they are processed.

Amazon SNS:- It is similar in that it is used to send out msgs to services, but it can also send out notifications to end users.
It does this in a different way called a publish/subscribe.
Ex:- Youtube channel subscribe → we get notifications.
(A channel for msgs to be delivered); SMS, email.

Monolithic Applications & Microservices:- (In this Approach to Application Architecture, if a single component fails, other ll fails or entire App fails)

Applications are made of multiple components. The components communicate with each other to transmit data, fulfill requests & keep the application running.
Suppose that u have an application with tightly coupled components. These components might include databases, servers, the user interface, business logic, etc.. This type of architecture can be considered → Monolithic App.
Microservices:- To help maintain application availability when a single component fails, we can design ~~your~~ our application through a microservice approach.

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⑥ Additional Compute Services:- AWS offers multiple Serverless Compute options.

Serverless means that you can't actually see or access the underlying infrastructure or instances that are hosting your Application.
Instead, all the management of the underlying environment from a provisioning, Scaling, high availability & maintainence perspective are taken care of for you. All you need to do is focus on your application & the rest is taken care of. AWS lambda is one Serverless Compute option.

It is a service that allows you to upload your code into what's called a Lambda fn.
Configure a trigger & from there, the service waits for the trigger. When the trigger is detected, the code is automatically run in a managed environment.
(It is more suited for Quick processing like a web backend, handling reqs or a backend expense report processing service).

→ If u still want the efficiency & portability => go through
=> AWS container services
Amazon Elastic CS.
Amazon ECS or Amazon Elastic kubernetes Services Amazon (EKS)

(Both of these services are container orchestration tools)

AWS Fargate :- It is a serverless compute platform for ECS or EKS.

Compute Services:
If you want
=> Host Traditional Apps
=> Full access to the OS.
=> Use Amazon EC2.

If you looking to host
short runnings fns.
→ Service oriented apps
→ Event driven apps.
→ No provisioning or managing servers
(AWS lambda)

for Docker Container based workloads on AWS
→ Amazon ECS or EKS.
→ platforms ⇒ Amazon EC2 that u manage
or in a serverless environment ⇒ AWS Fargate

Module 3 :- Global Infrastructure & Reliability :- (Co Many branch offices)

⇒ It's not good enough to have 1 giant data center + all of the resources are.
If smthg happen to that data center, like a power outage / Natural disaster ⇒ Everyone's Applications would go down all at once.
⇒ we need high availability & fault tolerance.
Turns out, it's not even good enough to have 2 giant data centers
Instead, AWS operates in all sorts of different areas around the world called Regions. (Course)

AWS Regions :- Inside each region, we have multiple data centers that have all the Compute, storage & other services we need to run Applications.
→ Each Region can be Connected to each other Region through a high speed fiber network, controlled by AWS, a truly global operation from Corner to Corner if u need it to be.
⇒ Any data stored in a particular region will be stayed in that region unless we explicitly, with the right Credentials & permissions, request the data to be exported.
⇒ Regional data sovereignty is part of the Critical design of AWS Regions. With data being sub to the local laws & statues of the Country the Region lives.

(12)

There are 4 business factors that go into choosing a Region:-

i) Compliance :- Before any of the other factors, u must look at your compliance requirements.

Do u have a requirement ? your data must live in Uk boundaries? Then u should choose the London Region, full stop.

None of the rest of the options matter. Most governances, though are not governed by such strict regulations.

So if u don't have a Compliance or regulatory control that dictates ur region, then u can look at the other factors.

~~ii) Proximity~~

ii) Proximity :- How close u are to ur customer base is a major factor because speed of light, still the law of the Universe.

Latency :- The time it takes for data to be sent & received.

iii) Feature Availability :- Sometimes, the closest region may not have all the AWS features u want. We constantly innovating on behalf of our customers. Every year, AWS releases 1000s of new features & products specifically to answer customer requests & needs. But sometimes those brand new services take a lot of new physical hardware that AWS has to build out to make the service operational & Sometimes that means we have to build the service out one Region at a time.

iv) Pricing :- Even when the hardware is equal to 1 Region to the next, some locations are just more expensive to operate in. Prices can be determined by the many factors, so AWS has very transparent granular pricing.

Availability Zones :-

AWS has data centers, lots of Data Centers, all around the world & each Region is made up of multiple Data Centers. AWS calls a single data center or a group of data centers → an Availability Zone or AZ.

Each AZ is one/more discrete data centers with redundant power, networking & connectivity.

When u launch an Amazon EC2 instance, it launches a virtual machine on a physical Hardware that is installed in an Availability Zone.

This means each AWS Region consists of multiple Isolated & physically seperates the AZs within a geographic Region. But we don't build AZ right next to each other because if a large scale incident were to occur, like a Natural Disaster then we could lose the Connectivity to everything in that AZ

Distance Ekkuva Vundali (Data Centers) ℓi

Recommendation :-
Run across at least 2 AZs in a Region

Edge locations :- (EL)

Caching Copies of data closer to the customers all around the world uses the Concept of Content delivery networks / CDNs.

CDNs are Commonly used & on AWS
We call our CDN Amazon cloudfront (ACF)
ACF → Service that helps deliver data, Videos, Applications & APIs to Customers around the world with low latency & High speed transfers.

ACF uses Edge locations, all around the world to help accelerate Communication with users, no matter where they are.

EL are seperate from Regions, so u can push content from inside a Region to a Collection of Edge locs around the world, in order to accelerate Communication & Content delivery.
AWS EL, also runs more than just ACF. They run a domain name service (DNS) known as Amazon Route 53, helping direct customers to the Correct web locs with incredibly low latency.

AWS outposts :- AWS will basically install a fully operational mini Region, right inside our own data center. (That is owned & operated by AWS, using same AWS infra, but isolated within our own building.)

→ AWS Edge locs run ACF to help get content closer to your customers, No matter where they are in the world.

AWS outposts :- Allows us to run AWS infrastructure right in our own data center.

Provision of AWS Resources :- APIs → Application Programming Interface

⇒ There are pre determined ways for us to interact with AWS services & u can invoke / call these APIs to provision, configure & manage ur AWS resources.

(14) Ex :- U can launch an EC2 instance.

Interacting with AWS Services :-

→ AWS Management console.
→ AWS CLI (Command line Interface).
→ AWS Software Development kit (SDKs)
→ Various other tools → AWS Cloudformation

to create requests to send to AWS APIs to create & manage AWS Resources.

AWS Management Console ⇒ * Test environments, View AWS Bills, View monitoring
work with Non-Tech resources.

(Browser based Through the Console).

AWS CLI :- It allows to make API calls using terminal on ur Machine.

Commands to launch / run.

Scripts.

SDKs → Allow us to interact with AWS resources through various programming languages.

(It makes easy for developers to create programs that use AWS without using the low level APIs.)

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AWS Elastic Beanstalk (EBS):- Service that helps us to provision Amazon EC2 based environments. Instead of clicking around the Console or writing multiple cmds to build out our network, EC2 instances, Scaling, ELBs.

we can instead provide our App code & desired configurations to the AWS EBS which then takes that info & builds out our envir for you.

AWS EBS makes it easy to save environment configurations, so they can be deployed again easily. AWS EBS gives convience of not having to provision & manage all of these pieces seperately, while still giving u the visibility & control of underlying resources. Focus on App Not Infra

AWS cloud formation :- Infrastructure as a code tool that allows u to define a wide variety of AWS resources in a declarative way using JSON/YAML text based docs. → called cloudform templates

It defines us what we want & supports many diff AWS resources, from databases, storage, ML etc.

It manages all the backend AWS API calls

Module 4 :- Networking :-

Amazon VPC (Virtual Private cloud) → A VPC lets u provision a logically isolated section of the AWS cloud + u can launch AWS resources in a virtual network that u define. These resources can be public facing so they have access to the internet, or private with no internet access, usually for backend services like databases/Application servers.

The public & private grouping of resources → Subnets. & they are ranges of IP Addresses in our VPC.

Connectivity to AWS :- VPCs essentially our own private network in AWS.

A VPC allows us to define our private IP range for our AWS resources, & us place things like EC2 instances & ELBs inside of our VPC.

Subnets are chunks of IP addresses in our VPC that allow us to group resources together.

Public traffic :- (Internet Gateway is used to connect a VPC to the Internet).

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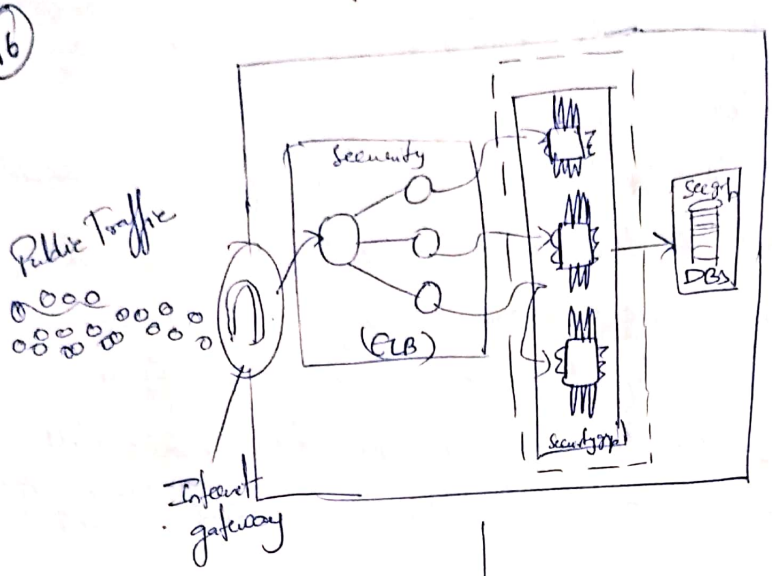
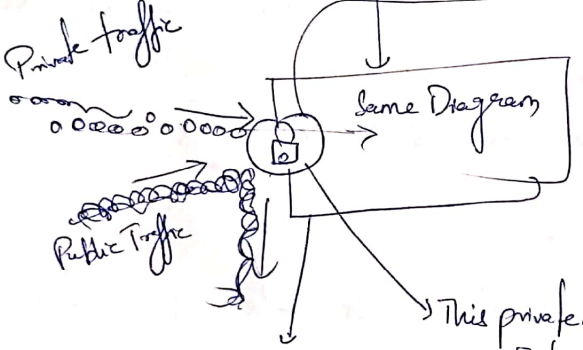


In order to allow traffic from the public internet to flow into & out of our VPC, u must attach what is called → Internet Gateway (IGW) (Vidhi Thakur)

IGW is like a doorway i.e open to the public.

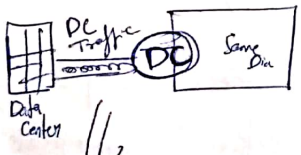
Private Traffic :-



We want the private gateway that allows only people in if they are coming from an approved network Not the public Internet.

This private doorway is known as Virtual private gateway.

& it allows you to create a VPN Connection b/w a private network.



(Lowest latency & Highest amount of Security)

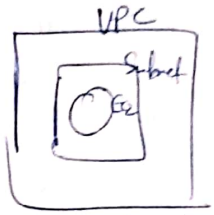
AWS Direct Connect :- DC allows us to establish a completely private, dedicated fiber connection from our data center to AWS. You work with a DC partner in our area to establish this connection. AWS DC → Provide physical line that connects our network to our AWS VPC.

AWS DC → It can be used to establish a private dedicated connection b/w a company's data center & AWS.

Subnets & Network Access Control Lists (ACLs)

AWS has a wide range of tools that cover every layer of security:—

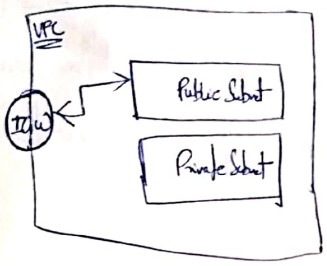
- → Network Hardening
- → Application Security
- → User Identity
- → Authentication & Authorization.
- → Distributed Denial of Service prevention.
- → Data Integrity.
- → Encryption.

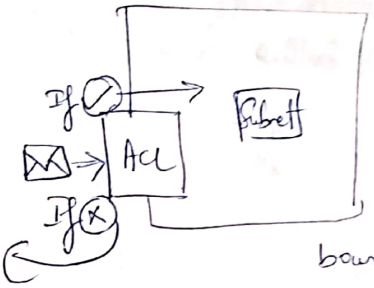


VPC → Internet gateway | EC2 → Sec grps
Subnet → ACL

Network Hardening :—

→ Packets





Subnets can also control Traffic permissions

Packets are messages from the Internet, every packet that crosses the subnet boundaries gets checked against something called ACLs.

→ It check is to see if the packet has permissions to either leave/enter the subnet based on who it was sent from & how it's trying to communicate.

ACLs → Passport Control officers.

ACL → evaluate a packet only if it crosses a subnet boundary (in/out).

It doesn't evaluate if a packet can reach a specific EC2 instance or not.

→ Sometimes, we will have the multiple EC2 instances in the same subnets, but they might have diff rules around who can send them msgs, what port those msgs are allowed to be sent to. So u need instance level network security as well.

→ We introduce → Security grps.

Every EC2 instance, when it's launched, automatically comes with a security grp. & By default, the security grp doesn't allow any traffic into the instance at all. (All ports are blocked) All IP addresses sending packets → blocked.

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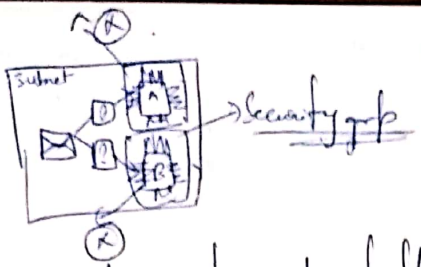
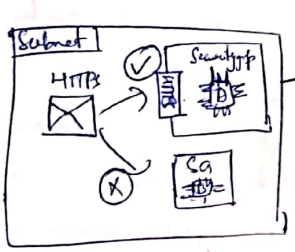


If u want an Instance to actually accept traffic from the outside, like say a message from a front end Instance / message from the Internet.

So obviously, u can modify ⇒ Security grp to accept a specific type of traffic.

In the case of website, u want web based traffic / HTTPS to be accepted but not other types of traffic, saying OS / Administration reqs. If NACLs are a passport control, a security grp is like the doorman of ur building, the building being the EC2 Instance, in this case.

The doorman will check a list to ensure that someone is allowed to enter the building but won't bother check the list on the way out. With security grps, u allow specific traffic in & by default, all traffic is allowed out.



→ only HTTPS allowed among (X)

ACL → Stateless & allows all inbound & outbound traffic.

Security grp	(Network ACL)
→ Stateful & deny all inbound traffic by default	→ Stateless (which remembers nothing) → (it always checks its list (own list)
(It has some kind of memory when it comes to who to allow in/out)	(Checks every single packet that crosses its border regardless of any circumstances)

Ex:- Let's start with Instance A. We wanna send a packet from Instance A to B in a diff subnet, same VPC, diff subnets. So (Ins A) sends the packet. Now, the 1st thing that happens is that packet meets the boundary of the Sec grp of Ins A. By default, all outbound traffic → allowed from a Sec grp.

The packet made it past the Security grp of Instance A. (Now it has to leave the Subnet boundary.)

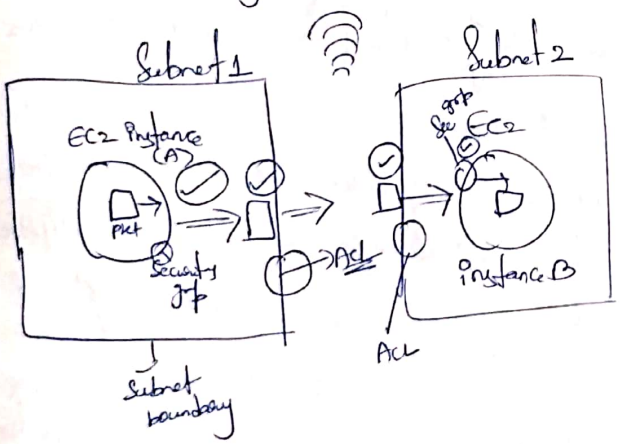
At the boundary, the passport must now make it through passport control, the network ACL. The ACL doesn't care about ~~the~~ what the security grp allowed. It has its own list of who can pass & who can't.

If the target address → allowed → U can keep going on ur journey, which it is.

Now, we have exited the original subnet - 1, now the packet goes to the target subnet - 2 where Instance B lives.

Now at this target subnet, once again, we have the passport control. Just because the packet was allowed out of its home country doesn't mean that it can enter the destination country / subnet in this case.

They both have unique passport officiers with their own checklists. U have to be approved on both lists, or u could get turned away at the border. Well, turns out the packet is on the approved list for this subnet, so it's allowed to enter through the network ACL into the subnet.



Now, we are approaching the target instance, Ins B.

Every EC2 instance has their own sec grp.

You wanna enter this instance, the doorman will need to check to see if u r allowed in, we are on the list.

The packet has reached the target instance. Once the transaction is complete, Now it's just time to come Home. It's return traffic pattern.

& Vice versa in reverse.

Global Networking:-

→ Amazon Route 53 (DNS) → Domain Name Service.
→ It's Highly available & Scalable.

DNS → Translation service. But instead of translating b/w languages, it translates website names into IP addresses that computers can read.

For Ex:- When we enter a website address into our browser, it contacts Route 53 to obtain the IP address of the site, say 192.1.1.1, then it routes us Computer/browser to that address.

Amazon Route 53 routing policies :—

→ Latency based routing.
→ Geolocation DNS.
→ Geoproximity
→ Weighted Round Robin

If we take geolocation DNS, that means we direct traffic based on the customers is located.
→ Ex :- So traffic from North America is routed to the Oregon Region.

(20)

We can use Route 53 to register the Domain Names, so we can buy & manage our own domain names right on AWS.
Speaking of websites, there is another service which can help ACF that speed up delivery of website assets to customers.

Module 5 :— Storage & Databases :-

Instance Stores & Amazon Elastic Block Store :—

Block level Storage : U can think of Block level storage as a place to store files.
A file being a series of bytes that are stored in blocks on disc.
When a file is updated, the whole series of blocks are not all overwritten.
Instead, it updates just the pieces that change. This makes it an efficient storage type when working with applications like databases, enterprise software or file systems.
When u use ur laptop or PC, you are accessing block-level storage. All block level storage is in this case is ur hard drive.
EC2 instances have hard drives as well & there are two diff types.

When u launch an EC2 instance, depending on the type of the EC2 instance u launched, it might provide u with local storage called Instance store volumes.
These vol are physically attached to the host, your EC2 instances running on top of.
& u can write to it just like a normal hard drive.
The catch here is that since this vol is attached to the underlying physical host, if u stop or terminate ur EC2 instance, all data written to the instance store vol will be deleted.

the reason for this, is that if u start an instance from a stop state, it's likely that EC2 instance will start up on another host. A host = that vol doesn't exist. Remember, EC2 instances are virtual machines & therefore the underlying host can change after stopping & starting an instance. Because of this ephemeral & temporary nature of instance store volumes, they are useful in situations if u can lose the data being written to the device such as temporary files, scratch data & data that can be easily recreated without consequences.

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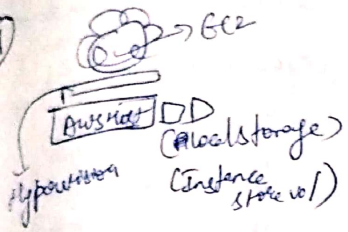
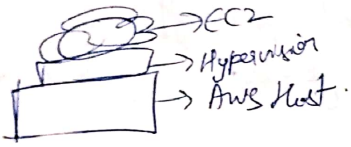


if EC2 gets terminated
then local storage will be vanished/deleted



Amazon Elastic Block Store (EBS) :- With EBS, we can create Virtual Hard Drives, that we call EBS volumes, that u can attach to ur EC2 instances. These are seperate drives from the local instance store volumes, & they aren't tied directly to the host & so u are easy is to running on.

This means, that the data that u write to an EBS volume can persists b/w stops & the starts of an EC2 instance.

EBS volumes come in all diff sizes & types. ⟶

How this works, is u define :-
→ Size, type & configurations of vol u need.

Provision the vols & then attach it ur EC2 instance. From there, u can configure ur application to write to the vol & you're good to go. If u stop & then start the EC2 instance, the data in the vol remains. Since the use case for EBS vol is to have a hard drive that is persistent, that ur applications can write to.

It's probably imp that u back that data up.

EBS allows u to take incremental backups of ur data → Snapshots.

It's very imp that u take Snapshots of ur EBS volumes. This way, if a drive even becomes corrupted, u haven't lost ur data & u can restore that data from a snapshot.

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Amazon S3 (Simple Storage Service & Bucket)

Amazon S3 :- It allows us to store & retrieve an unlimited amount of data at any scale. Data is stored as objects but instead of storing them in a file directory u can store them in a ~~[illegible]~~ buckets.

Think of a file sitting on our hard Drive, that is an object & think of a file directory → that is the Bucket.

Store data as obj in Buckets 5TB max	version obj create multiple buckets

The max object size → 5TB.

You can also version objs to protect them from accidental deletion of an object. What this means is that u always retain the previous versions of objects as like a paper trail. U can even create multiple buckets & store them across diff classes or tiers of data. U can then create permissions to limit who can see or even access objects. & u can even stage data b/w different tiers. These tiers offer mechanisms for different storage use cases such as data that needs to be retained for several years.

S3 standard → Comes with 11 9's of Durability. (AWS can sustain the concurrent loss of data in 2 separate storage facility)

That means an obj stored in S3 standard has 99.999999999% of Durability.

(11 9's)

→ Data is stored in such a way that AWS can sustain the concurrent loss of data in 2 separate storage facilities.

→ This is because data is stored in at least 3 facilities.

So multiple copies reside across locs.

Another way of useful to use S3 → static website Hosting.

→ Collection of HTML files

& u can simply upload all ur HTML, static web assets, & so forth into a bucket & then checking a box to host it as static website. U can enter the bucket URL → Instant website & we say static.

Amazon S3 Standard Infrequent Access (S3-Standard-IA) (S3-IA) | Which is used for data that is accessed less frequently but requires rapid access when needed. It means perfect place to store backups, disaster recovery files

Amazon S3 Glacier :- If I need to archive the data.
(S3 G)

To use Glacier, u can simply move data to it, or u can create vaults & then populate them with archives
u can employ an S3 Glacier vault lock policy & lock a vault.

You can specify controls such as write once/read many or WORM, in a vault lock policy & lock the policy from future edits. Once locked, the policy cannot be changed

Amazon S3 Lifecycle Mngmt :- Move data automatically b/w tiers.

Tib = 1.1TB

Amazon EBS :-
weighing sizes up to
→ Size ⇒ 16 TiB
→ Secure termination of their EBS instance.
→ Solid state by default.
→ HDD options. (Spinning platters)

S3
→ Unlimited storage
→ Individual objs → 5TBs
→ WORM.
→ 99.999999999% Durable.

(For updates → EBS)

S3 :- (Images)
→ web Enabled
→ Regionally Distributed
→ offers cost Savings
→ Serverless

Obj storage
If treats any files as complete discrete obj
(Change → Re-upload)

Block Storage :-
Breaks the files into small component parts/Blocks.
(updates the blocks).

If u are using complete objs :-
occassional changes → S3
If u are doing complex read, right, change fns → EBS

Amazon EFS (Elastic File System) :- EFS → managed file system
Extremely common for businesses to have shared file systems across their apps.

Ex :- U might have multiple servers running analytics on large amount of data being stored in a shared file system. This data traditionally has been hosted on premises.
In this on pre, u would have to ensure that the storage u have can keep up with the amount of data that u r storing. Make sure backups are taken & the data is stored redundantly as well as manage all the servers hosting that data.

With EFS, u can keep existing file systems in place. (24)
→ EFS:- Multiple Instances can access the data in EFS at the same time.
It scales up & down as needed.

AMAZON EBS :-

EBS volumes store data within a single AZs
EFS multiple AZs.

→ Volumes attach to EC2 instances.
→ Availability zone level resource.
→ Need to be in the same AZ to attach EC2 instances.
→ u can save files on it / run a Database / store Applications.
(Hard Drive)
→ Volumes do not automatically scale to give u more storage.

EFS :-

→ Multiple Instances reading & writing simultaneously.
→ Linux file System
→ Regional resource.
→ Automatically scales.

Amazon RDS → storing data in an Amazon Aurora Database & using SQL to organize data.

Amazon RDS (Relational Database System) (service RDBMS) :- It means we store data in a way such that it relates to other pieces of data.
(Ex :- If we had a customer entry/record, we store that in Customer Table).

(SQL) → Structured Querying Language)
Database Sys → MySQL, PostgreSQL, Oracle, MicrosoftSQL etc.. (needs configuration)

But is there a way to easily move them to the cloud? → Yes.
We can do Lift & Shift & migrate ur Database (DB) to run on Amazon EC2.

⇒ The other option for running your DBs in the cloud is to use a more managed service called Amazon RDS. (It supports all the major DBs Engine).
→ It includes :- Automated patching, Backups, Redundancy, failover, Disaster recovery.
(→ gets backup for S3) (→ Point in Recovery)

Amazon Aurora :- (Most managed Relational Database Option) 2 forms → MySQL, → PostgreSQL
(upto 15 Rep) (Data Replication) (1/10th cost of Commercial DBs)

Amazon DynamoDB:- (Serverless Database)

→ It is a Non-Relational, No-SQL Database. (Simple flexible Schemas / Not complex)
→ It is purpose built ⇒ it has a specific use cases & it is not the best fit for every (Dataset) workload out there.
→ It has millisecond response time (Highly scalable)
→ It is fully managed & it is highly scalable.

(With DynamoDB, u create tables. A DynamoDB table is just a place where u can store & query data.) Data is organised into items, & items have attributes.
Attributes are just diff features of data.
If u have 1 item in ur table or 2 Million items in ur table. DynamoDB manages the underlying storage for u. (Need not to worry about the scaling of the system).
DynamoDB stores this data redundantly across AZs.

Amazon RDS :-

→ Automatic high availability, recovery provided.
→ Customer ownership of Data & Schema. (Control)
→ Customer control of network.

Amazon DynamoDB :-

→ The NoSQL using key-value pair ⇒ No advanced Schema.
Able to operate Global Database.
→ Massive throughput capabilities.
→ PB (Petabyte) Scale potential.
→ Granular API access.

Amazon Redshift:- (ARS) (used to query & Analyze data across a data warehouse)

Data Warehouses → They are engineered specifically for this kind of Big data, if u are looking at historical analytics as opposed to operational analysis.

→ Historical may be as soon as :- Show me last hour's sales nos across all the stores. The key thing is, the data is now set.

(26)

ARS ⇒ It is data warehousing as a Service.

⇒ Massively Scalable.

RS (Nodes in multiple petabyte sizes ⇒ very common).

In fact, in coperation with Amazon RS spectrum, u can directly run a single SQL query against exabytes of unstructured data running in data lakes.

⇒ But it's more than just being able to handle massively larger data sets.

⇒ RS uses a variety of innovations that allow u to achieve up to 10 times higher performances than traditional DBs, when it comes to business intelligence workloads.

→ RS allows u to get started with a single API call.

AWS Database Migration Service (AWS DMS):-

AWS DMS ⇒ enables us to migrate relational databases, Non-Relational DBs & other types of data stores onto AWS in a secure & easy fashion.

→ The source DataBase (DB) remains fully operational during the migration.

⇒ Downtime is minimized for applications that rely on that database.

→ The source & target databases don't have to be of the same type.

(Databases that are of same type (source/target) ⇒ Homogenous Migrations) & can be from MySQL to Amazon RDS for MySQL; Microsoft SQL Server to Amazon RDS for SQL Server. The process is fairly straightforward since schema structures, datatypes & database code is compatible b/w source & Target.

The Source can be on premises, Amazon EC2, Amazon RDS

The target can be a Database in Amazon EC2, Amazon RDS.

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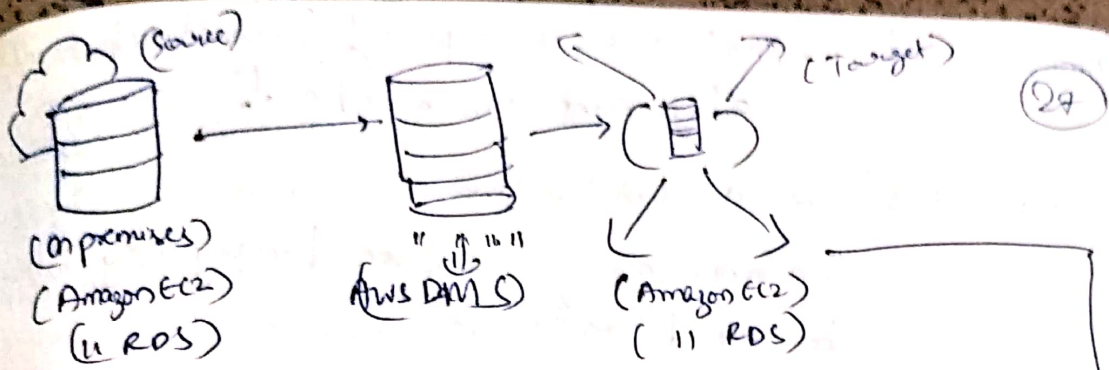
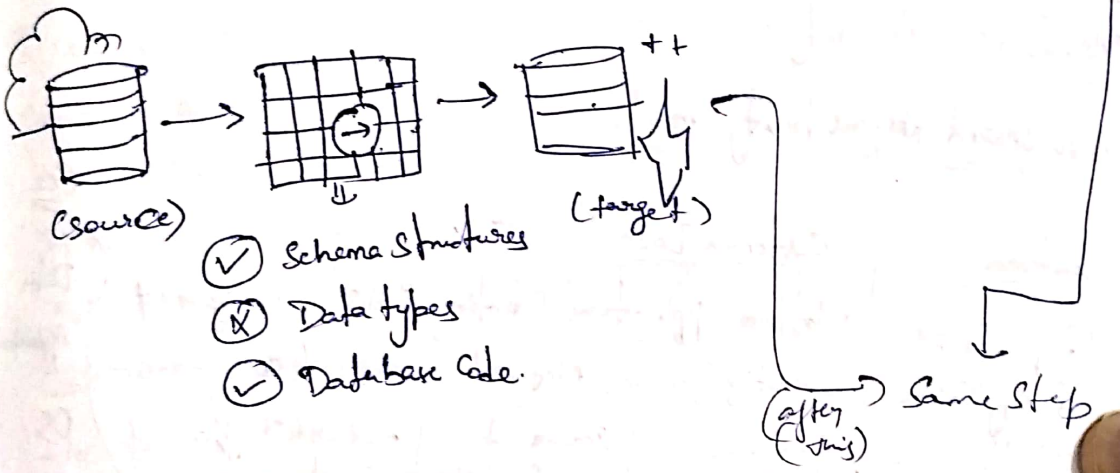


2nd Type → Heterogeneous Migrations when Source & Target Databases are of different types. (2 Step process)



But these are not the only use case for DMS

Others include :—

→ Development & test Database Migrations. → when u want to develop this to test against production data but without affecting production users.

→ Database Consolidation. (Is when u have several DBs & want to consolidate them in 1 central DB).

→ Continuous Database replication.

↓

(Is when u use DMS to perform cts Data Replication.)

(This could be for disaster recovery or because geographic seperation).

Additional Database Services:— There is no one size fits all database for all purposes.

Amazon DocumentDB → Great for content management, catalogs, user profiles.

Amazon Neptune → A graph database, engineered for social networking & recommendation engines, also great for fraud detection needs.

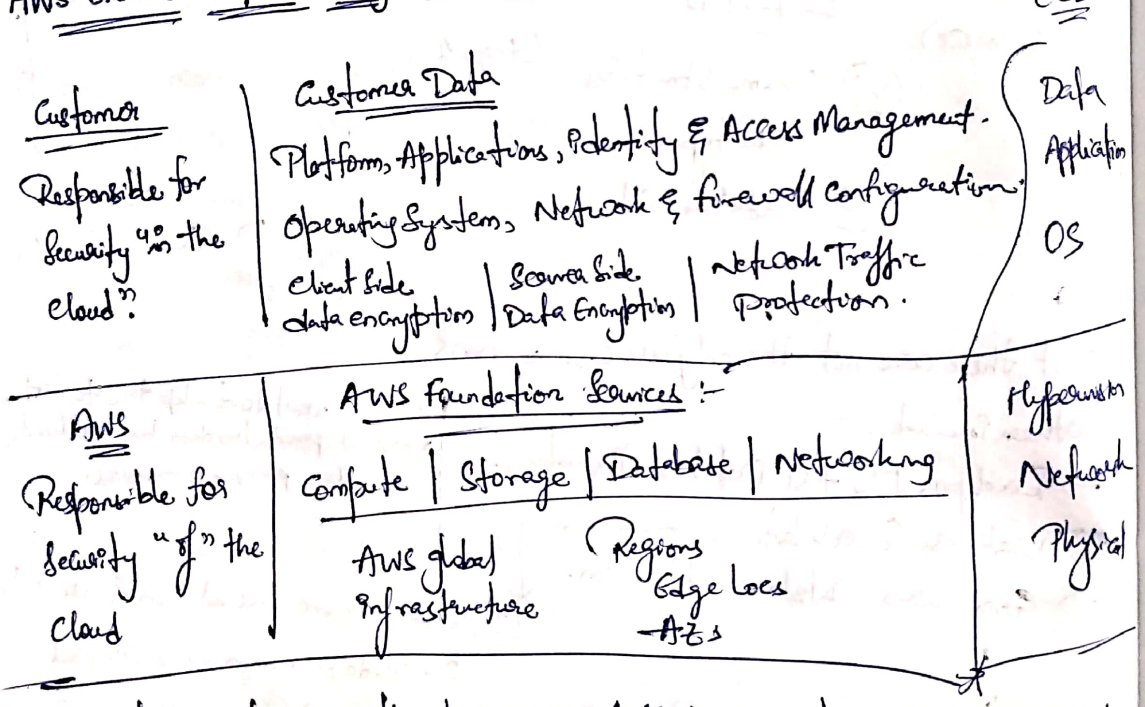
Amazon Managed Blockchain :- It solves the part of the Equation, but adds a huge decentralization component.

Amazon QLDB / Quantum Ledger DataBase
=> An immutable systems of record + any entry can never be removed from the Audits.

DAX → The DynamoDB Accelerator (A native caching layer designed to dramatically improve read times for our non-relational data).

Module 6 :- Security :-

AWS Shared Responsibility Model :-



=> When it comes to securing the business on AWS => The Customer & AWS are both responsible for the Security. (Patching Software on Amazon EC2 instances
Setting permissions for a S3 obj)
(Customer side)

Ex :- The Builder Constructed the House with 4 walls & a door.
It's their responsibility to make sure the walls are strong & the doors are Solid. (AWS)

→ It's our responsibility the homeowner, to close & lock the doors. (Customer)

User permission & Access :— (IAM policy:- Doc that grants/denies permissions to AWS services & resources)

→ When u create an AWS Account, u are given what is called the root acc user & has permission to do anything they want inside of that acc. (29)

→ AWS acc root user :- Access & Control any resource in the acc.
we can spin up EC2 instances, databases, blockchain services or literally anything. Because that user is so powerful.

→ You turn on Multi factor Authentication (MFA) :- To ensure that u need not only the email & password but also a randomized token to log in.

→ You control access in a granular way by using the AWS service, AWS Identity & Access Management (IAM).
u can create IAM users. When u create an IAM user, by default, it has no permissions. The user can't even log into the AWS acc at 1st, it has absolutely zero permissions. It can't launch EC2 instance, can't create S3 Bucket, Nothing. You have to explicitly give the user permission to do anything in that acc.
By default, all actions are denied. u have to explicitly allow any action done by any user. u give people access only what they need & nothing else
↳ known as Least privilege principle.

→ The way that u grant/deny permission is to associate what is called an IAM policy to an IAM user.

→ An IAM policy is a JSON doc that describes what API calls a user can & can't make.

⇒ Effect : Allow/Deny, Action: Any AWS API Call.
Resource : which AWS resource the API call is for.

→ 1 way to make it easier to manage ur users & their permissions is to organize them into IAM groups.

AWS IAM :- Root user, Users, Groups, Policies, Roles.

⇒ u can create identities in AWS that are called Roles. → Access-to-temporary permissions
⇒ Roles :- Associated permission that allow/deny specific actions & Assumed for temporary amount of time. (Similar to user, but no username & password)

AWS Organizations :-

1 way to install order & to enforce who is allowed to perform certain things in what account is to make use of an AWS service called AWS Organizations.

The easiest way to think of organizations is as a central location to manage multiple AWS accs.

U can manage billing control, access, compliance, security & share resources across ur AWS accs.

=> Centralized Management.
=> Consolidated Billing
=> Hierarchical groupings of accs.
=> AWS services & API actions access control.

Service Control policies (SCPs) :- to specify the maximum permissions for member accs in the organization.

In essence, with SCPs, u can restrict which AWS services, resources & individual API actions, the users & roles in each mem acc can access)

An Organizational unit (OU) & an individual member acc (SCP can supply)

Compliance :- The act of obeying a law/rule, especially one that controls a particular industry or type of work. (Inspection)

=> GDPR (General Data Protection Regulation) :- If u run software that deals with consumer data in EU, u would need to make sure that u are in compliance with GDPR. (shared responsibility).

AWS Artifact :- 1 place we can access the documents is through a service called AWS Artifact.

=> With AWS Artifact, u can gain access to compliance reports done by 3rd parties who have validated a wide range of compliance standards.

AWS Artifact tasks :-

Access AWS compliance reports on demand.
Review, Accept & manage agreements with AWS.

Denial of Service Attacks :-

Distributed Denial of Service Attacks (DDoS) :-

It's an attack on an enterprise's Infrastructure.

The objective of a DDoS attack is to shut down an app's ability to function by overwhelming the system to the point it can longer operate.

→ In Normal operations, an app takes requests from clients & returns results

→ In DDoS, the bad actor tries to overwhelm the capacity of an app, basically to deny anyone the services.

⇒ Problem: UDP flood (weather) → Solution :- Security groups (Only allow in particular traffic) (operates at the AWS network level)

→ Problem :- Slow loris Attack (queue) like → Solution: ELB → Elastic load Balancer.

⇒ For the sharpest, most sophisticated attacks, AWS also offers specialized defense tools → AWS Shield with AWS WAF.

⇒ AWS WAF uses a webapp firewall to filter incoming traffic for the signatures of bad actors. It has extensive ML capabilities, can recognize new threats as they evolve & proactively help defend our systems against an ever-growing list of destructive vectors.

Additional Security ~~goes~~ Services :-

→ Encryption :- Securing a msg/data in a way that only authorized parties can access it.

→ Encryption at Rest :- At rest, we mean our data is idle (just sorted & not moving).

→ Encryption at Transit :- It means the data is traveling b/w service & client. (accessing the service)

→ AWS KMS (Key Management Service) :- Enables to perform encryption operations through the use of cryptographic keys.

⇒ Amazon Inspector → Helps to improve security & compliance of ur AWS deployed apps by running an automated security assessment against ur Infrastructure.

Specifically, it helps to check on deviations of security best practices.

↳ It consists of 3 parts:-

i) Network Configuration reachability piece
ii) Amazon agent → Can be installed at ec2 instances.
iii) Security assessment services

⇒ Amazon Guard Duty ← Another security dimension in our threat detection offering

(Analyzes ctg streams of metadata generated from ur acc & network activity.)

Found on Amazon CloudTrail, VPC Flow logs, DNS logs

→ uses Integrated Threat Intelligence → malicious IP addresses, anomaly detection & ML to identify threats more accurately

(It runs independently from ur other AWS services).

An employee requires temporary access to create several Amazon S3 buckets

→ IAM role is best choice for task.
(u can assume to gain temporary access to permissions)

→ MFA → Authentication process that provides an extra layer of protection for ur AWS Acc.

→ The root user → The identity that is established when u 1st create an AWS Acc.

↳ AWS Shield service helps protect ur apps against DDos attacks.

Module-7 :- Monitoring & Analytics :-

Monitoring :- The Idea of observing systems, collecting metrics, evaluating those metrics over time & then using them to make decision / take actions.

→ It's imp to set up mon in cloud.

→ It will help u measure how our systems are performing, alert u when things are n't right, & even help u debug & troubleshoot issues.

Amazon Cloudwatch :- It allows u to monitor ur AWS infrastructure & Amazon CW :- the apps u run on AWS in real time.

→ It works by tracking & monitoring metrics.

→ Think of metrics as variables that are tied to ur resources.

→ CW allows us to create a custom metric called Espresso count & once it hits 100, it alert staff to clean.

→ [CW Alarm] → u set a threshhold for a metric, & when that threshold is reached cloudwatch can generate an alert & trigger an action.

→ CW integrated with SNS.

→ Access all ur metrics from a central loc.

→ Gain visibilibility into ur apps, infrastructure & services.

→ Reduce MTTR & improve TCO.

→ Drive Insights to optimize apps & operational resources.

Amazon CloudTrail (CT) :- The principle is simple :- Trust but verify.

ACT records API Calls for ur Acc. The recorded info includes the identity of the API Caller, the time of the API Call, the source IP address of the API caller & more. CT as trail of breadcrumbs (log of actions) that someone has left them.

Every request gets logged in the CT Engine.

(The Engine records exactly who made the request)

Tasks of Amazon Cloud Trail:—

→ Track user Activities & API requests throughout ur AWS infrastructure.
→ Filter logs to assists with operational analysis & troubleshooting.

AWS Trusted Advisor:— AWS has an automated advisor known as AWS Trusted advisor (AWSTA)

→ When running a business, u might need some advisors who can come in from the outside & advices u..

AWS TA :— This is a service that u can use in ur AWS Acc that will evaluate ur resources against 5 pillars.

Those pillars :—

i) Cost optimization
ii) Performance
iii) Security
iv) Fault Tolerance
v) Service Limits.

Trust Advisor is real time ~~time~~ runs through a ~~check~~ series of checks for each pillar in ur Acc, based on AWS best practices & it Compiles categorized items for u to look into, & u can view them directly in the AWS Console.

Actions of Amazon Cloudwatch :—

→ Monitor ur resources utilization & Performance.
→ Access metrics from a single dashboard.

AWS Trusted Advisor service enables u to review the security of Amazon S3 buckets by checking for open access permissions.

Amazon Guard Duty → Service that provides intelligent threat detection for ur AWS environment & resources. It identifies threats by ctsly monitoring the network activity & acc behavior within ur AWS environment.

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Module 8:- Pricing & Support:- AWS provides a lot of free tools to help u plan & analyze ur budgets for ur AWS environments.

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AWS free tier:-

The free tier offers 3 different ways to try out a service :-

i) Always Free :- Means a service is available to all AWS Customers & it does nt expire.

ii) 12 Months Free (1 Year) :- Mean u have 12 months to try out a service following ur Initial sign up date to AWS. So the date u Created ur AWS Acc.

iii) Trails :- Some services offer a short term free trial & then they expire after that period ends.

Ex:- Always free ⇒ AWS Lambda

12 Months free ⇒ Amazon S3 free for 1 year for up to 5GB of Standard Storage.

Trails → AWS Lightsail offers 1 month trial up of 750 hrs of usage.

AWS free tier Services:-

→ Amazon SageMaker.
→ " Comprehand Medical.
→ " DynamoDB.
→ " SNS
→ " Cognito & etc.

AWS Pricing Concepts:-

⇒ How AWS pricing works:- It offers a range of cloud computing services with pay-as-go-pricing.

⇒ Pay for what u use?

For each services, u pay for exactly the amount of services that u actually use, without requiring long term Contracts & Complex licensing.

Pay less when u reserve :- Some services offer reservation options that provide a significant discount compared to On-Demand Instance pricing.

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Pay less with volume based discounts when u use more :- Some services offer tiered pricing, so the per unit cost is ↓ lower with ↑ed usage. (Incrementally) (36)

Ex :- S3 storage space wise, the less u pay for it per GB.

AWS pricing calculator :- AWS pricing Calc explore AWS services & create an estimate for the cost of ur use cases on AWS. U can organise ur AWS estimates by grps that u define. A grp can reflect how ur company is organized, such as providing estimates by cost centers.

→ When u have created an estimate, u can save it & generate a link to share it with others.

Ex :- AWS lambda Pricing, Amazon EC2 Pricing
(Pay less when u reserve) (u pay only the compute time)

Amazon S3 pricing
↳ Storage
↳ Requests & Data Retrievals
↳ Data Transfer
↳ Management & Replication.

Billing Dashboard :- Use the AWS Billing & Cost Management dashboard to pay ur AWS bill, Monitor ur usage & Analyse & Control ur costs.

→ Compare ur current month to date balance with previous month & get a forecast of the next month based on current usage.

→ View month to date spend by service.

→ U Free tier Usage U U .

→ Access Cost explorer & Create budgets.

→ Purchase & Manage Savings plans.

→ Publish AWS Cost & usage Reports.

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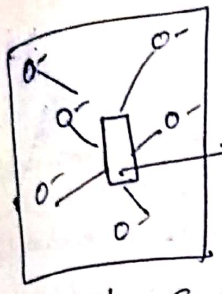
Consolidated Billing :-



→ Anni bills main organization ke velthundhi rather than on individuals.

A singular Company will more than likely have multiple AWS accs & u can manage multiple aces in AWS by using AWS orgs.

→ Consolidated Billing :- At the end of every months, instead of having to pay an AWS bill for every sigle acc, u can roll those bills up into 1 bill owned by the owner of the org. It makes it easier to keep track of ur bills.

→ u don't get 100 bills, if u have 100 AWS accs.

They would make 1 bill for the billing cycle.

So, its easier for the company to manage.

- Consolidated billing
  - => Simplifies billing process
  - => Share savings across accs.
  - => Does it cost u any extra money. (Free feature).

AWS Budgets :- u can create budgets to plan ur source usage, service costs & Instance reservations. The Info in AWS budgets updates 3 Times a day. This helps u to accurately determine how close ur usage is to ur budgeted amounts or to the AWS free tier limits.

→ u can also set custom alerts when ur usage exceeds the budgeted amount. (Mobile Balance)

AWS Cost Explorer :- A tool that enables u to visualize, understand & manage ur AWS costs & usage over time. (Gives ur historical data) (Tracks)

It includes a default report of the costs & usage for ur top 5 cost accounts AWS services. You can apply custom filters & grps to analyze ur data.

AWS Support Plans :-

It offers 4 different support plans to help u troubleshoot issues, lower costs & efficiently use AWS services!

→ Basic → free for all AWS customers. (AWS Trusted Advisor) It includes access to whitepapers, documentation & support communities.
→ Developer → Best practice use, client side diagnostic tools, Building Hub architecture
→ Business → All AWS Trusted Advisor checks, identify AWS offerings, limited support for 3rd party
→ Enterprise → A Technical Account manager, Application architecture guidance, Infrastructure event management.

Basic Support :-

→ 24/7 Customer Service
→ Documentation
→ Whitepapers
→ Support Forums
→ AWS Trusted Advisor
→ Free for all

Developer

→ Basic Support
→ Email access to Customer support

Business

→ Basic & Developer support.
→ AWS Trusted Advisor provides full set of best practice checks
→ Direct phone access to cloud support engineers.
→ Infrastructure event management.

Business Support plan (Cost)

Enterprise :-

Enterprise includes all AWS Trusted Advisor checks at the lowest cost.

→ All previous tiers support
→ 15 min SLA for business critical workloads.
→ Technical Ace Manager (TAM). → They specialize not only in proactively monitoring ur environment & assisting with optimizations but provide Infra event management, well Architected reviews & operation reviews.

5 pillars of Well Architected framework :-

→ Operational Excellence
→ Security
→ Reliability
→ Performance efficiency
→ Cost optimization

TAMs work with clients to review architectures the using the well A framework.

AWS Marketplace :- It is a curated digital catalog that streamlines the steps to find, deploy & manage 3rd party software running in our AWS Architecture. (37)
(Helps in accelerating Innovation).

→ 1 click deployment → that allows them to quickly procure & use products from 1000s of Software Sellers.

(Pay as yo options)

Categories :- Infrastructure Products, Business Apps, Data Products & Devops.

Enterprise focused features :-

→ Custom terms & pricing
→ A private Marketplace.
→ Integration to our procurement systems.
→ cost management tools.

Module 9 :- Migration & Innovation :-

AWS cloud Adoption framework (AWS CAF) :- Migrating to the cloud is a process.
The AWS professional services team has created something called the AWS CAF that can help us manage the process through guidance.
The cloud Adoption framework exists to provide advice to our company to enable a quick & smooth migration to AWS.
The Framework organizes guidances into 6 areas :- (Perspectives)

→ Business
→ People
→ Governance
} Focus on the Business Capabilities.

→ Platform
→ Security
→ Operations
} Focus on the Technical Capabilities

AWS CAF Action Plan :- Helps guide our Organization for Cloud migration

Each perspective is used to uncover gaps in our skills & processes, which are then recorded as inputs. These inputs are then used as the basis for creating

→ Platform perspective of the AWS Cloud Adoption framework helps u design, implement, & optimize ur AWS infrastructure based on ur business goals & perspectives.

(40)

→ The Platform Perspective of the AWS CAF also includes principles for implementing new solutions & migrating on-premises workloads to the cloud.

→ The Business perspective helps u to move from a model that separates business & IT strategies into a business model that integrates IT strategy.

→ The operational perspective focuses on operating & recovering IT workloads to meet the requirements of ur business stakeholders.

→ The people perspective helps HR employees prepare their teams for cloud Adoption by updating organizational processes & staff skills to include cloud based competencies.

Migration Strategies :-

The 6 R's :- U can decide which option among the 6 R's is the best fit based on time, cost, priority, criticality.

→ Rehosting ⇒ This is otherwise known as lift & shift & easy thing for business to do because of not making any changes.
→ Easier to optimize apps once they live in the cloud
→ Because the org has the better skills to do.

→ Replatforming ⇒ Lift, tinker / shift.
It is purely lift & shift but instead of pure 1 to 1, u might make few cloud optimizations.
(Not touching any core code in the process)

→ Retire ⇒ Some parts of enterprise IT portfolio are just no longer needed
(It can save significant cost & effort of a team)
(without being used)

→ Retain ⇒ Some apps are about to be deprecated but maybe not just yet for 3 to 8 months.
(Should only migrate what makes sense for the business)

→ Repurchase :- This is common for companies looking to abandon legacy software vendors & get a fresh start as a part of migration. (11)

→ Refactoring :- U are writing new code. → driven by strong business need to add features / performance that might not be possible to perform ; but now within our reach.
(Dramatic changes to our architecture can be very beneficial to our enterprise) but this will come at the highest potential cost in terms of planning & human effort.

→ Repurchasing migration strategy involves moving to a different product.
(If replaces an existing app with a cloud based version).

→ Refactoring → changing how an App is architected & developed, typically by using cloud native features.

→ Replatforming :- Selectively optimized aspects of an app to achieve benefits in the cloud without changing the core architecture of the app. (Lift, tinker & shift).

AWS Snow family :- Collection of physical devices that help to physically transport up to exabytes of data into & out of AWS.

→ SnowCone, Snowball, SnowMobile.

→ SnowCone → Small, rugged & secure edge computing & data transfer device
features → 2 CPUs, 4 GB of memory, 8TB of usage storage.

→ Snowball
→ 2 types of devices :-
→ Edge storage optimized (80TB storage capacity) (Suited for large-scale data migrations & recurring transfer workflows, in addition to local computing with higher capacity needs)
→ Edge Compute optimized → (powerful computing resources for use cases : ML, Data analytics etc)

→ SnowMobile :-
✓ Exabyte-scale data transfer service used to move large amounts of data to AWS.
100TB Storage Capacity

Innovation with AWS:-

→ Amazon Sagemaker → Service enables us to quickly builds, train & deploy ML models.

→ Amazon Textract → ML service that automatically extracts text & data from scanned docs.

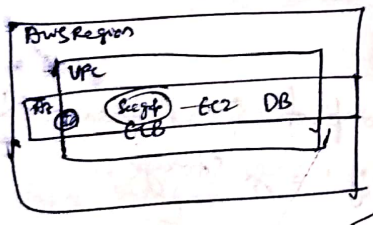
→ Amazon Lex → Service that enables u to build conversational interfaces using voice & text.

→ AWS Deepracer → Autonomous 1/18 scale race car that u can use to test reinforcement learning models.

→ Security perspective of the AWS CAF helps u structure the selection & implementation of permissions.

→ Governance perspective helps u to identify & implement best practices for IT governance & support business processes with technology.

Module 10 :- The Cloud Journey :-



→ AWS well Architected framework

⇒ Operational excellence
⇒ Security
⇒ Reliability
⇒ Performance efficiency
⇒ Cost optimization.

(Capability to run & monitor systems to deliver business value & to continually improve supporting processes)

(Helps u to understand how to design & operate reliable, secure, efficient & cost effective sys in AWS Cloud. It provides a way for u to consistently measure ur architecture against best prac & design principles & Identify areas for improvement).

→ Reliability focuses on the ability of a workload to consistently & correctly perform its intended fns.

→ ~~Effic~~ Performance efficiency focuses on using computing resources efficiently to meet sys requirements & to maintain that efficiency as demand changes & technologies solve.

→ Security → protecting data, systems & assets & using cloud technologies to improve the security of our workloads.